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(1) Extension Program/ Project Title: SMART WASTE: EMPOWERING COASTAL COMMUNITIES ON WASTE MANAGEMENT	
Proponent Name:	Princess Lydia S. Fuentes
Academic Rank:	Associate Professor I
Co-Authors (with academic rank):	JOCELYN S. LEGASPI- Asso. Prof IV JOSELITO C. SITJAR, Asso. Prof V RIO L. VERGABERA – Instr. III ABNER F. OROFEO- Instr. I
Campus/Satellite College/Address:	Dayao Satellite College
Telephone/Cell Phone/Fax Number:	
E-mail Address:	plsfuentes@capsu.edu.ph

(2) Category (please check) ☐ Education and Humanities ☐ Agriculture/Fisheries/Livelihood ☒ Health/Environment ☐ Engineering and Technology

(3) Background:  The proposed project, SMART WASTE: EMPOWERING COASTAL COMMUNITIES ON WASTE MANAGEMENT is designed to address the problems and challenges on waste management of the coastal communities of Roxas City that led to pollution, more particularly, on plastic pollution and other wastes. The worldwide issue of waste management has been so long that arises from various economic sectors, human activities, people consumption of goods and services, and personal habits. It has social and economic impacts from the local to the international level. Roxas City is known as the country's "seafood capital" due to its abundant fish and other seafood production encompasses of freshwater and saltwater fish, molluscan shellfish, cephalopod mollusks and crustaceans. There are 23 coastal barangays in Roxas City that contributed to 11.1 percent of total share of Western Visayas to the country's total value of fish production making the second contributor in the country (BFAR, 2019). Despite the industry's potentials, research shows that plastic pollution may affect the socio-economic aspect of the coastal communities (Fuentes, PL, 2021). This effect is one of the result of poor waste management of the coastal	
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communities. Thus, this project is proposed to empower coastal communities and create a locally contextualized waste management and utilization plan.

(4) RATIONALE (WHY should the project be implemented? **WHAT need** is the project trying to address? – how the problem is affecting the targeted beneficiaries, previous efforts done and status, any legal bases and related literature supporting the proposed project, link the relevance of the project to development priorities like mandate and development goals, the comparative advantage to handle the project, and **WHY** act now and not later)

Waste has become one of the most serious environmental issues that the world is dealing with. The accumulation of waste such as plastic and other products in the environment lead to pollution which imposes a hazardous effect on wildlife and human food chain (Ganguly, 2018), human health, groundwater, land and air. Data on plastics in municipal solid waste from 1960-2018 reported by the American Chemistry Council and the National Association for PET Container Resources, revealed that generation of plastics in 2018 was 35.7 million tons in the United States, which was 12.2 percent of Municipal Solid Waste generation wherein the total amount of plastics combusted was 5.6 million tons. In 2018, landfills received 27 million tons of plastic which was 18.5 percent of all Municipal Solid Waste landfilled. In 2021, the Philippines estimated to emit 35% of the ocean's plastic (Lourens JJ Meijer, et al, 2021). Roxas City had notable government-led programs to address the issues on waste management such as the City Ordinance No. 075-2001, or the Comprehensive Solid Waste Management System that established a material recovery facility which includes solid waste transfer station or sorting station, drop-off center, a composting facility, and a recycling facility. However, in 2017, it was found out that the Roxas City sanitary landfill located in one of the coastal barangays that dumping of mixed-wastes wreak havoc into the landfill's functional capacity that resulted to its early destruction. With this, the city passed the 2017 Waste Segregation and Sanitary Landfill Protection Act of Roxas City that prohibits the intentional littering or disposal of residual waste at household or establishment source. The said ordinance sets punishment and penalties amounting from ₱ 3,000.00 to ₱ 5,000.00 plus cancellation of business permit if violator is a business operator. In addition, the implementation of City Ordinance No. 030-2011 "No Plastic Ordinance" prohibits business establishments or individuals from selling or providing plastic bags as packing materials to consumers. In January 2011 the "No Plastic Days" were observed every Monday and Wednesday. Information banners and streamers on the day of observance of the said ordinance were displayed on various strategic places to remind the public on the said initiative. The move of the city government also promotes its "Bring Your Own Bag" campaign. Based on the ordinance, individual violators will be meted with ₱ 500 and ₱ 1,000 for first and second offenses, respectively, while a third violation will penalize an individual with ₱ 2,000 and or imprisonment of not more than six months upon the discretion of the court. However, the implementation and sustainability of the different policies were not realized. Thus, this project is forwarded to revisit the waste management practices and strengthen the efforts using input-process-product design.



This project tends to address **SDG-12** environmental issues on waste management particularly on **substantially reduce waste generation** and **circular economy**. The current extension project is also aligned to the CAPSU RDE Agenda 2023-2027 on **Climate Change Adaptation and Mitigation, Sustainable Community, Coastal Management' and Policy Formulation**.

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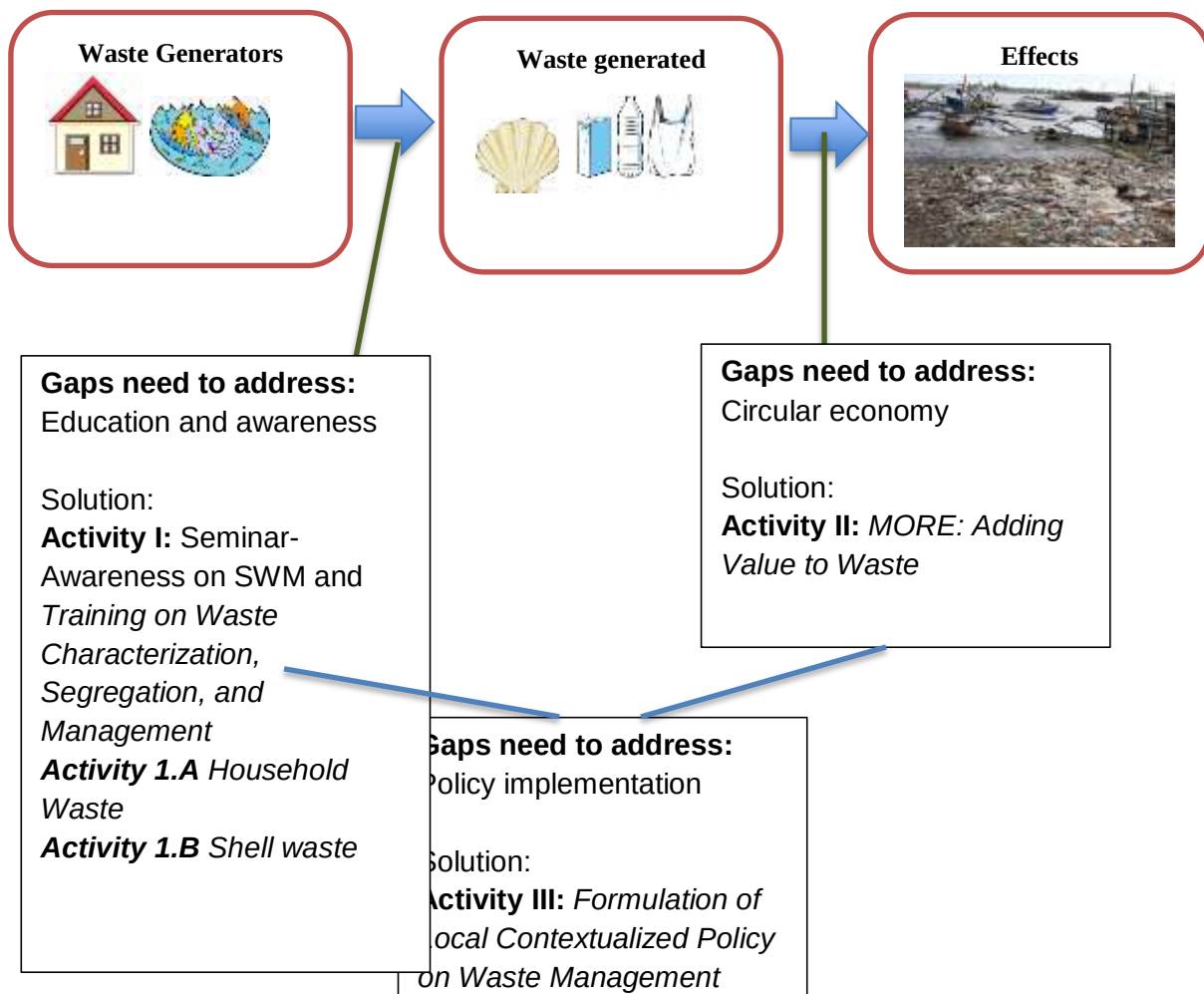
(5) GOALS AND OBJECTIVES

The main objective of this extension project is to empower coastal communities and create a locally contextualized waste management and utilization plan. Specifically, it aims to:

1. to increase level of awareness among coastal communities through the conduct of seminar on RA 9003 Solid Waste Management;
2. to characterize and segregate solid waste through the conduct trainings on waste characterization, segregation, and management on;
 - a. household waste; and
 - b. shell waste
3. to convert waste into valuable functional items by the conduct of trainings on recycling and upcycling of waste; and
4. to formulate contextualized local policy for coastal communities on waste management and utilization.

(6) PROJECT DESCRIPTION (what are the **features** of the project process/tools used in the project ID, location, duration, beneficiaries, results chain/framework, project cycle – [needs/gaps identified, set of solutions noted to address the needs])

The project overview is described below.



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(7) PROJECT IMPLEMENTATION AND Monitoring & Evaluation SYSTEM (how will the project be implemented, how will the project be tested if it is working, when - detailed activities to be undertaken by component to generate the outputs, support the description of activities through a matrix or work plan identifying how each activity will be delivered, who will be responsible for WHAT, WHEN, WHERE – state your Activities, Work Plan - Project Schedule supported with a Gantt chart, Project Management [proponent, partners and their roles and responsibilities including counter parting scheme, if any, capacity to handle the project, and accompany with the simple management structure. Provide theoretical background and apply it to the project. Describe how the project will be monitored and evaluated during the planning/pre-implementation, implementation, and operation/post-implementation phases. M&E framework must be accompanied by a **DATA Collection Plan**)

“BEFORE IMPLEMENTATION”

OBJECTIVE	To conduct an expectation survey among the beneficiaries.
TASKS	- Conduct a utilization forum/briefing among barangay officials, BFARMC, and residences. - Do an expectation STUDY survey base on the objectives of the project. - Execute MOA
PERSONS INVOLVED	<i>Facilitators:</i> Project proponents <i>Participants:</i> Barangay Officials, BFARMC representative, residents
TIME FRAME	June-July 2024
DATA COLLECTION PLAN	Conduct an interview/ FGD among participants.
OUTPUT	- A baseline data of extension expectations of participants. - Notarized MOA

“DURING IMPLEMENTATION” – Activity I: Seminar-Awareness on SWM and Training on Waste Characterization, Segregation, and Management

OBJECTIVE	1. increase level of awareness among coastal communities on RA 9003 2. to characterize and segregate solid waste
TASKS	Seminar on the Implementation of RA 9003 Seminar-Training on Waste Characterization, Segregation, and Management Establishment of Material Recovery Facilities
PERSONS INVOLVED	<i>Resource Persons:</i> Experts in Criminal Justice Education, SWM Advocates <i>Facilitators:</i> Project proponents <i>Participants:</i> Barangay Officials, BFARMC, residents, barangay tanods, waste collector unit
TIME FRAME	August- October 2024
DATA COLLECTION PLAN	- Conduct pre-test & post-test on the level of awareness of the participants on SWM Implementation - Conduct field observation and interview on SWM Implementation
OUTPUT	Conducted Seminar-Training Quantitative data on the Level of Awareness on SWM Qualitative data on the implementation of SWM Established MRF

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“DURING IMPLEMENTATION” – Activity II: MORE: Adding Value to Waste

OBJECTIVE	1. convert waste into valuable functional items by the conduct of trainings on recycling and upcycling of waste; 2. produce valuable products from waste materials
TASKS	• Training on Waste Valorization (recycling and upcycling)
PERSONS INVOLVED	<i>Resource Persons:</i> Experts in Arts and Crafts, Waste Technology <i>Facilitators:</i> Project proponents <i>Participants:</i> Barangay Officials, BFARMC, residents
TIME FRAME	November 2024 –January 2025
DATA COLLECTION PLAN	- Output inventory - Conduct field observation and interview
OUTPUT	Conducted Training on Waste Valorization Produced valuable products from waste materials

“DURING IMPLEMENTATION” – Activity III: Formulation of Local Contextualized Policy on Waste Management

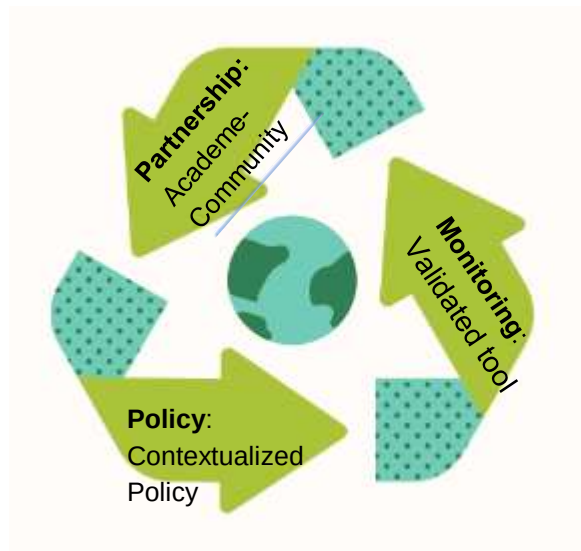
OBJECTIVE	1. formulate of local context policy and sustainability plan
TASKS	• Formulation of Policy and Sustainability Plan
PERSONS INVOLVED	<i>Facilitators:</i> Project proponents <i>Participants:</i> Barangay Officials, BFARMC, residents
TIME FRAME	February-March 2025
DATA COLLECTION PLAN	Conduct field observation and interview on the implementation of the policy
OUTPUT	Approved local context policy and sustainability by the host institution and local communities.

“AFTER IMPLEMENTATION”

OBJECTIVE	1. determine the increase in level of awareness on RA 9003 among coastal communities; 2. conduct debriefing session/ culmination activity with public exhibits on the SMART WASTE project just like recycled/upcycled items, crafts, and other utilities, policy plan, etc, as results/products derived from the project;
TASKS	Culmination activity with public exhibits on the SMART WASTE project just like recycled/upcycled items, crafts, and other utilities, policy plan, etc, as results/products derived from the
PERSONS INVOLVED	<i>Facilitators:</i> Project proponents <i>Participants:</i> Barangay Officials, BFARMC, residents
TIME FRAME	April- May 2025
DATA COLLECTION PLAN	- Conduct a rapid impact study - Document success stories
OUTPUT	- A short video clip/documentation on the impact of the project to the community. - A qualitative research, through semi-structured interview and field-observation on the impact of the project to the beneficiaries after a period of time.

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(8.) SUSTAINABILITY PLAN (All projects have limited time durations. Can the project be supported after its project life? How will the organization sustain the project in terms of dedicated staff? How will the beneficiaries continue to carry out the activities after the project? Is there any financial support required for the project after the project has ended?)



The project will be implemented for approximately 2 years to explore its effectiveness to the beneficiaries. More than people or pool of experts, successful project should involve partnerships and coalitions at their core. In this project, it refers to the partnerships between CAPSU and the local communities. Strengthening partnerships have the greatest potential to make an impact on coastal communities. Building synergy between organizations with common goals has the potential to increase visibility of the project and stress the importance of the message of waste management. Second, the integration of the project on the existing systems and political support among local communities are important for sustainability. Lastly, monitoring and evaluation and sharing of results is important to evaluate the outcomes and effectiveness of the project, ensuring that resources are allocated strategically to different initiatives.

(9.) RESOURCE REQUIREMENTS - (a. **Direct Costs** – (Implementing team – manpower (who – names and positions, how many?, Facilities and Equipment needed, Supplies and Materials, Travel, Hauling, Etc) b. **Indirect costs** (rental, electricity, administrative cost 5% to 15%) c. Contingencies (price increases, unforeseen and emergency expenses) 5% of the direct cost, **Budget**)

Budgetary Requirement

Counterpart	Other	TOTAL
	Funding	Funding

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I. Personal Services

Activity I. Salaries/ Honorarium	P 30, 000.00
Activity II. Salaries/ Honorarium	P 30, 000.00
Activity III. Salaries/ Honorarium	P 30, 000.00

II. Maintenance and Other Operating Expenses

A. Direct Cost

Travel Expenses (Local) P 15, 000. 00

Supplies and Materials

Material Recovery Facility P70, 000.00

Training Materials P30,000.00

Facility and Equipment Rental Expenses P 10, 000.00

Documentation Printing, Photocopying Expenses P 15, 000.00

B. Indirect Cost

Utilities/Miscellaneous P 15, 000

GRAND TOTAL P 245, 000.00

(10) Executive Summary (A significant part of the proposal. Summarize the project so that readers can rapidly become acquainted with the problem, background information, implementation scheme, and resource requirements. Intended as an aid to decision-making since it covers all sections, but do not copy and paste and write in the same order as the main document.)

The proposed project, SMART WASTE: EMPOWERING COASTAL COMMUNITIES ON WASTE MANAGEMENT is designed to address the problems and challenges on waste management of the coastal communities of Roxas City that led to pollution, more particularly, on plastic pollution and other wastes. The main objective of this extension project is to empower coastal communities and create a locally contextualized waste management policy and utilization plan. The present extension project is divided into three phases/projects: Training on Waste Characterization, Segregation, and Management; Project MORE: Adding Value to Waste; and Formulation of Local Contextualized Policy on Waste Management.

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